
PHYSICAL CHEMISTRY(CHE201)

FINAL CURRENT PAPER FALL 2022(8March2023)

SHORT QUESTIONS(3)

From Midterm syllabus

Thermodynamics

Thermodynamics: (Greek words for “thermo=heat” and “dynamics=power”) is the study of heat, work, energy, and the changes they produce in the states of systems.

Thermodynamics studies the relationships between the macroscopic properties of a system.

A key property in thermodynamics is temperature.

Thermodynamics is sometimes defined as *the study of the relation of temperature to the macroscopic properties of matter*.

Thermodynamic Systems:

- The macroscopic part of the universe under study in thermodynamics is called the **System**

The parts of the universe that can interact with the system are called the **surroundings**.

○ TYPES OF SYSTEM

- 1. Open system** is one where transfer of matter between system and surroundings can occur.
- 2. Closed system** is one where no transfer of matter can occur between system and surroundings.
- 3. Isolated system** is one that does not interact in any way with its surroundings. • An isolated system is obviously a closed system, but not every closed system is isolated.

Condition for which non isolated system stay in equilibrium state:

A non-isolated system is in equilibrium when the following two conditions hold

A. The system's macroscopic properties remain constant with time

B. Removal of the system from contact with its surroundings causes no change in the properties of the system

- If condition (A) holds but (B) does not hold, the system is in a steady state.

Irreversible Process:

An irreversible process is one in which heat is transferred through a finite temperature. In summary, processes that are not reversible are called irreversible.

Examples: • Relative motion with friction • Combustion • Diffusion • Free expansion • Throttling • Electricity flow through a resistance • Heat transfer • Plastic deformation.

Phase:

If a system is not homogeneous, it may consist of a number of homogeneous parts. A homogeneous part of a system is called a phase.

A system composed of two or more phases is heterogeneous.

Distinguish between colligative properties:

We distinguish between four colligative properties:

vapor pressure lowering,
freezing point depression,
boiling point elevation,
and osmotic pressure.

Molar mass & Atomic mass:

❖ **Atomic Mass**

The average mass of an atom or molecule is called the **atomic mass** or the molecular mass.

Molecular masses are commonly expressed in units of atomic mass units (amu)

where 1 amu is one-twelfth the mass of a ^{12}C atom

❖ **Molar Mass**

The mass per mole of a pure substance is called its **molar mass M**. For example, for H_2O , $M = 18.015 \text{ g/mole}$. The molar mass of substance i is

$$M_i = m_i / n_i$$

where m_i is the mass of substance i in a sample and n_i is the number of moles of i in the sample. The molar mass M_i and the molecular weight $M_{r,i}$ of i are related by $M_i = M_{r,i} \times 1 \text{ g/mole}$, where $M_{r,i}$ is a dimensionless number.

From Final Syllabus

Define the following:

- **Syrup:** A solution of water and sugar is called syrup.
- **Brine:** A solution of sodium chloride (common table salt) in water is called brine.
- **Saline :** A sterilized specific concentration (0.15 molar) of sodium chloride in water is called saline
- **Seltzer:** A solution of carbon dioxide in water is called seltzer.

Three points of Maxwell distribution/contributions:

- Careful studies of gases show that the distribution is not a bell or normal distribution, but a Maxwell- Boltzmann distribution.
- A sketch of a Maxwell distribution is given here.
 - 1) The peaks are not symmetrical.
 - 2) There are more molecules at higher speed than at lower speeds.
 - 3) When the temperature increases, the peak shifts to the right.

Properties of Colloids:

- The particles of dispersant are between about 500 nm to 1 nm in diameter.
- The mixture does not separate on standing in a standard gravity condition. (one 'g').
- The mixture does not separate by common filter, but might be filterable by materials with a smaller mesh.
- The mixture is not necessarily completely homogeneous, but usually close to being so.
- The mixture may appear cloudy or almost totally transparent, but if you shine a light beam through it, the pathway of the light is visible from any angle. This scattering of light is called the Tyndall effect.
- There usually is not a definite, sharp saturation point at which no more dispersant can be taken by the dispersing agent.
- The dispersant can be coagulated, or separated by clumping the dispersant particles with heat or an increase in the concentration of ionic particles in solution into the mixture.

- There is usually only small effect of any of the colligative properties due to the dispersant.

Mechanism of chain reaction:

Elementary steps used for mechanisms of chain reactions can be grouped into the following categories:

- Initiation step
 - Chain propagation steps
 - Chain branching steps
 - Chain inhibition steps
 - Chain termination steps
- For example, the chlorination of ethane is a chain reaction, and its mechanism is explained in the following way.
 - If we mix chlorine, Cl_2 , and ethane, CH_3CH_3 , together at room temperature, there is no detectable reaction.
 - However, when the mixture is exposed to light, the reaction suddenly initiates, and explodes.

Effect of Catalyst:

“A catalyst provides a reaction route with a lower activation energy”.

☐ ☐ A catalyst is a substance that participates in a chemical reaction and increases the reaction rate without undergoing a net chemical change itself.

☐ ☐ Consider, for example, the decomposition of hydrogen peroxide in the presence and absence of different catalysts.

☐ ☐ Because most catalysts are highly selective, they often determine the product of a reaction by accelerating only one of several possible reactions that could occur.

☐ ☐ Most of the bulk chemicals produced in industry are formed with catalyzed reactions.

☐ ☐ Recent estimates indicate that about 30% of the gross national product of the United States and other industrialized nations relies either directly or indirectly on the use of catalysts.

Pseudo First Order Reaction:

To avoid more complicated, expensive experiments and calculations, we can use the pseudo first order reaction, which involves treating a second order reaction like a first order reaction.

The functional form of the decay kinetics is similar to the first order kinetics and the system is said to operate under pseudo first order kinetics.

- In a pseudo first order reaction, we can manipulate the initial concentrations of the reactants.
- One of the reactants, A, for example, would have a significantly high concentration, while the other reactant, B, would have a significantly low concentration.
- We can then assume that reactant A concentration effectively remains constant during the reaction because its consumption is so small that the change in concentration becomes negligible.

Solvation and example:

The interactions between the solute particles and the solvent molecules is called solvation. A solvated ion or molecule is surrounded by solvent.

Example: An important specific example of solvation is hydration, where the solvent is water. In general, the rule of like-attracts-like applies to solvation: Polar solutes such as sodium chloride are solvated by polar solvents such as water; they are not solvated by non-polar solvents such as benzene.

Solvent Effects:

The nature of the solvent can also affect the reaction rates of solute particles.

☐ ☐ For example, a sodium acetate solution reacts with methyl iodide in an exchange reaction to give methyl acetate and sodium iodide.

☐ ☐ This reaction occurs 10 million times more rapidly in the organic solvent dimethylformamide [(CH₃)₂NCHO] than it does in methanol (CH₃OH).

☐ ☐ Although both are organic solvents with similar dielectric constants (36.7 for DMF versus 32.6 for methanol), methanol is able to hydrogen bond with acetate ions, whereas

DMF cannot.

☐ ☐ Hydrogen bonding reduces the reactivity of the oxygen atoms in the acetate ion.

☐ ☐ Solvent viscosity is also important in determining reaction rates.

☐ ☐ In highly viscous solvents, dissolved particles diffuse much more slowly than in less viscous solvents and can collide less frequently per unit time.

☐ ☐ Thus the reaction rates of most reactions decrease rapidly with increasing solvent viscosity.

Relaxation method

- Many chemical reactions are complete in less than a few seconds, which makes the rate of reaction difficult to determine. In these cases, the relaxation methods can be used to determine the rate of the reaction.

- The term relaxation is used to describe a reaction's return to equilibrium.

- An equilibrium system is subjected to an external perturbation, such as temperature change.

- When the change is applied suddenly, the lagging time it takes the system to reach the new equilibrium position is related to the K_f and K_r constants and is called relaxation time, τ .

- There are three different techniques (each are different processes that suddenly disturbs a reaction), that are used to witness the relaxation time.

- ❖ These are

- 1) ☐ **Pressure Jump** (When a reaction is sensitive to changes in pressure, the resulting relaxation can be used to determine the rate constant of reaction).
- 2) ☐ **Temperature Jump** (When there is sudden change in temperature its called temperature Jump).
- 3) ☐ **Electric Field Jump** (The rate of reaction can be calculated by the sudden change in the electric field).

LONG QUESTIONS(5)

Mechanism of stopped flow method of concentration(S&L)/ methods of stopped flow/

The stopped-flow technique allows for the evaluation of solution-based kinetics on a milliseconds timescale with a very small volume of reactants used.

Mechanism:

☐ This technique involves two reactants held in separate reservoirs that are prevented from freely flowing by syringe pumps.

☐ The reaction is initiated by depressing the reactant syringes, and thus releasing the reactants into the connecting "mixing chamber" where the solutions are mixed.

☐ The reaction is monitored by observing the change in absorbance of the reaction solution as a function of time.

☐ As the reaction progresses it fills the "stop syringe" which then expands until it hits a block at the point when the reaction has reached a continuous flow rate, thereby

stopping the flow and the reaction, and thus allowing the researcher to calculate the exact initial rate of reaction.

☐ The Stopped Flow technique works because within milliseconds of combining the two reactants the absorbance can be read.

☐ In addition, the stop syringe assures for a steady rate of flow pas the spectrophotometer so that reactants are being added to solution and forming products at a consistent rate.

How stopped flow techniques work

- The stopped-flow technique allows for the evaluation of solution-based kinetics on a milliseconds timescale with a very small volume of reactants used.
- Imagine you wanted to compute the initial rate of a reaction, one that, when the reactants combined changed color or fluoresced, however the reaction was far too fast to detect via the human eye alone.
- How would you go about determining the reaction rate? Could you use the continuous flow technique? But what if you only had a limited volume of the reactants at your disposal?
- The continuous flow technique requires that you have enough of the reactants to constantly flow through the spectrophotometer at a continuous rate.
- In cases like this, the Stopped Flow technique is probably more appropriate.
- Stopped-Flow allows for the rate of a solution-based reaction to be determined in milliseconds, and with a very small volume of reactants.

Elementary steps for reaction mechanism:

☐ Based on numbers of molecules involved in the elementary step, there are three kinds of elementary steps:

- 1) ☐ Unimolecular step (or process),
- 2) ☐ Bimolecular step,
- 3) ☐ Trimolecular step.

1. Unimolecular Steps:

- *When a molecule or ion decomposes by itself, such an elementary step is called a unimolecular step (or process).*
- A unimolecular step is always a first order reaction.
- The following **examples** are given to illustrate this point:
- $\text{O}_3 \rightarrow \text{O}_2 + \text{O}$, $\text{Rate} = k[\text{O}_3]$
- or in general
- $\text{A} \rightarrow \text{B} + \text{C} + \text{D}$, $\text{Rate} = k[\text{A}]$
- $\text{A}^* \rightarrow \text{X} + \text{Y}$, $\text{Rate} = k[\text{A}^*]$
- A^* represents an excited molecule.

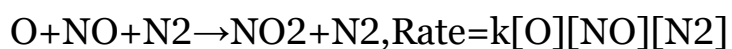
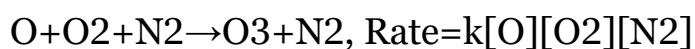
2. Bimolecular Step (process):

- ☐ *A bimolecular process involves two reacting molecules or ions.*
- ☐ The rates for these steps are 2nd order .
- ☐ **Examples:**
- ☐ $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$, $\text{Rate} = k[\text{NO}][\text{O}_3]$
- ☐ $\text{Cl} + \text{CH}_4 \rightarrow \text{HCl} + \text{CH}_3$, $\text{Rate} = k[\text{Cl}][\text{CH}_4]$
- ☐ In general:
- ☐ $\text{A} + \text{B} \rightarrow \text{X} + \text{Y}$, $\text{Rate} = k[\text{A}][\text{B}]$

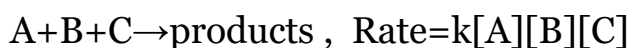
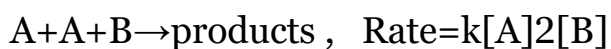
3. Trimolecular Steps(process):

□ A trimolecular process involves the collision of three molecules.

For example:



In general, trimolecular steps may be,



Azeotropes

- An azeotrope is a mixture that exhibits the same concentration in the vapor phase and the liquid phase.
- This is in contrast to ideal solutions with one component typically more volatile than the other; this is the reason we use distillation to separate materials.
- If the mixture forms an azeotrope, the vapor and the liquid concentrations are the same, which preventing separation via this approach.
- Azeotropes are a mixture of at least two different liquids.
- Their mixture can either have a higher boiling point than either of the components or they can have a lower boiling point.
- Azeotropes occur when fraction of the liquids cannot be altered by distillation.
- Typically when dealing with mixtures, components can be extracted out of solutions by means of Fractional Distillation, or essentially repeated distillation in stages (hence the idea of 'fractional').
- The more volatile component tends to vaporize and is collected separately while the least volatile component remains in the distillation container and ultimately, the result is two pure, separate solutions.

Limitations of Thermodynamics

❖ Limits of Thermodynamics in Reaction mechanisms 1:

Thermodynamics is a powerful approach toward understanding chemical reactions, but only provides part of the picture.

Specifically:

- 1) Thermodynamics only points the way
- 2) Thermodynamics says nothing about how long it takes to get there
- 3) The stoichiometric equation for the reaction says nothing about its mechanism

❖ Limits of Thermodynamics in Reaction mechanisms 2:

- Three reactions that look alike, but are different: Consider, for example, the gas-phase formation reactions of the hydrogen halides from the elements. The thermodynamics of these reactions are all similar (they are all highly exothermic), but their dynamics (their kinetics and mechanisms) could not be more different.

OR

Although thermodynamics is a very powerful tool, we must be aware of the limitations that exist to its application.

- It deals with the properties of macroscopic system.
- It does not provide mechanism of process.
- It does not consider internal structures of atoms and molecules.
- It does not deal with the rate of reaction.
- The most important applications of thermodynamics are based on one fact: all natural systems go to equilibrium unless an external force is applied.
- And thermodynamics tells us what the system's properties are at equilibrium (see the alternative definition of thermodynamics). In other words, thermodynamics tells us where the system is going. But it does NOT tell how to get to equilibrium,

or even if we can GET to equilibrium. To think of it in another way we could say that thermodynamics tells us we can go from New York to London. But it does not tell us that we cannot do so by walking.

- **Example:** A more concrete example is with carbon. Most people know that carbon can exist as either diamond or graphite. Thermodynamics tells us that the most stable form, at room temperature and normal atmospheric pressure is graphite. But of course when we buy a diamond it does appear to change. Actually, it is changing but at an extremely slow rate. It takes millions of years for it to happen.

Effects of enzyme on activation energy with example

- However, if a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed.
- Enzymes can be thought of as biological catalysts that lower activation energy.
- Enzymes are proteins or RNA molecules that provide alternate reaction pathways with lower activation energies than the original pathways
- Enzymes affect the rate of the reaction in both the forward and reverse directions; the reaction proceeds faster because less energy is required for molecules to react when they collide.
- Thus, the rate constant (k) increases.
- A catalyst helps lower the activation energy barrier, increasing the reaction rate.
- In the case of a biological reaction, when an enzyme (a form of catalyst) binds to a substrate, the activation energy necessary to overcome the barrier is lowered, increasing the rate of the reaction for both the forward and reverse reaction.

Practical applications of kinetic studies

- Kinetic studies are important in understanding reactions, and they also have practical implications.
- For example, in industry, reactions are conducted in reactors in which compounds are mixed together, possibly heated and stirred for a while, and then moved to the next phase of the process.
- It is important to know how long to hold the reaction at one stage before moving on, to make sure reaction has finished before starting the next one.

By understanding how a reaction takes place, many processes can be improved.

- For example, if it is known that a particular intermediate is involved in a reaction, the use of conditions (such as certain solvents) that are incompatible with that intermediate might be avoided.
- In addition, reagents might be added that would make certain steps in the reaction happen more easily.
- Not only are kinetic studies important in industry, but they are also used to understand biological processes, especially enzyme-catalyzed reactions.
- They also play a role in environmental and atmospheric chemistry, as part of an effort to understand a variety of issues ranging from the fate of prescription pharmaceuticals in wastewater to the cascade of reactions involved in the ozone cycle.

Factors That Affect Reaction Rates 1:

- Although a balanced chemical equation for a reaction describes the quantitative relationships between the amounts of reactants present and the amounts of products that can be formed, it gives us no information about whether or how fast a given reaction will occur.
- This information is obtained by studying the chemical kinetics of a reaction, which depend on various factors: reactant concentrations, temperature, physical states and surface areas of reactants, and solvent and catalyst properties if either are present.

- By studying the kinetics of a reaction, chemists gain insights into how to control reaction conditions to achieve a desired outcome.

Factors That Affect Reaction Rates 2:

Phase and Surface Area effects:

- When two reactants are in the same fluid phase, their particles collide more frequently than when one or both reactants are solids (or when they are in different fluids that do not mix).
- If the reactants are uniformly dispersed in a single homogeneous solution, then the number of collisions per unit time depends on concentration and temperature.
- If the reaction is heterogeneous, however, the reactants are in two different phases, and collisions between the reactants can occur only at interfaces between phases.
- The number of collisions between reactants per unit time is substantially reduced relative to the homogeneous case, and, hence, so is the reaction rate.
- The reaction rate of a heterogeneous reaction depends on the surface area of the more condensed phase
- Automobile engines use surface area effects to increase reaction rates.
- Gasoline is injected into each cylinder, where it combusts on ignition by a spark from the spark plug.
- The gasoline is injected in the form of microscopic droplets because in that form it has a much larger surface area and can burn much more rapidly than if it were fed into the cylinder as a stream.
- Similarly, a pile of finely divided flour burns slowly (or not at all), but spraying finely divided flour into a flame produces a vigorous reaction.

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